

ASSESSMENT TOOL-1

C programming output prediction is given as CO1 AT3

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Q1. Reverse an Array:

```
1  #include <stdio.h>
2
3  int main() {
4      int a[100], n, i;
5
6      printf("Enter number of elements: ");
7      scanf("%d", &n);
8
9      printf("Enter elements: ");
10     for(i = 0; i < n; i++)
11         scanf("%d", &a[i]);
12
13     printf("Reversed array: ");
14     for(i = n - 1; i >= 0; i--)
15         printf("%d ", a[i]);
16
17     return 0;
18 }
19
```

Output:

```
Enter number of elements: 5
Enter elements: 10 20 30 40 50
Reversed array: 50 40 30 20 10
```

Q2. Palindrome String:

```
1  #include <stdio.h>
2  #include <string.h>
3
4  int main() {
5      char str[100];
6      int i, len, flag = 1;
7
8      printf("Enter a string: ");
9      scanf("%s", str);
10
11     len = strlen(str);
12
13     for(i = 0; i < len / 2; i++) {
14         if(str[i] != str[len - i - 1]) {
15             flag = 0;
16             break;
17         }
18     }
19
20     if(flag)
21         printf("Palindrome");
22     else
23         printf("Not a Palindrome");
24
25     return 0;
26 }
```

Output:

```
Enter a string: madam
Palindrome
```

Q3. Searching an Element in a Sorted Array — Binary Search:

```
1  #include <stdio.h>
2
3  int main() {
4      int a[100], n, key, low, high, mid, i;
5
6      printf("Enter number of elements: ");
7      scanf("%d", &n);
8
9      printf("Enter sorted elements: ");
10     for(i = 0; i < n; i++)
11         scanf("%d", &a[i]);
12
13     printf("Enter element to search: ");
14     scanf("%d", &key);
15
16     low = 0;
17     high = n - 1;
18
19     while(low <= high) {
20         mid = (low + high) / 2;
21
22         if(a[mid] == key) {
23             printf("Element found at position %d", mid + 1);
24             return 0;
25         }
26         else if(a[mid] < key)
27             low = mid + 1;
28         else
29             high = mid - 1;
30     }
31
32     printf("Element not found");
33
34     return 0;
35 }
```

Output:

```
Enter number of elements: 5
Enter sorted elements: 10 20 30 40 50
Enter element to search: 30
Element found at position 3
```